AI-Based Autonomous Navigation and Real-Time Collision Avoidance System for Micro-Vehicles

MINI PROJECT REPORT

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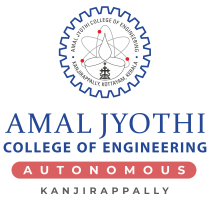
ELECTRONICS AND COMMUNICATION ENGINEERING

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DECLARATION

We undersigned hereby declare that the project report "**AI-Based Autonomous Navigation and Real-Time Collision Avoidance System for Micro-Vehicles**", submitted for partial fulfillment of the requirements for the award of degree of Minor in Electronics And Communication Engineering of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of **Mr. Binu Mathew**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

Autonomous micro-vehicles and intelligent ground robotics require robust, real-time perception and deterministic motion arbitration to navigate dynamic environments safely. Traditional robotic control systems frequently decouple high-level computer vision algorithms from low-level motor actuation, introducing severe latency bottlenecks and uncalibrated safety failures when deployed on resource-constrained embedded platforms. This project proposes an integrated, embedded Edge-AI autonomous navigation and real-time collision avoidance system implemented on a Raspberry Pi single-board computer driving an intelligent robotic testbed.

The system combines a low-latency vision pipeline using a native camera module running at 30 frames per second, adaptive Region of Interest (ROI) filtering, and centroid lane-tracking with an INT8-quantized YOLOv8 neural network executed through the LiteRT inference engine. To resolve complex indoor obstructions where semantic object detection alone is insufficient, a multi-sector corridor density scanner evaluates prospective clearance pathways in real time. Sensor fusion is established by coupling the optical perception stack with front and rear ultrasonic distance sensors through a calibrated voltage-divider interface. Motion execution is governed by a deterministic Finite State Machine (FSM) driving a dual H-bridge motor controller using 100 Hz software Pulse Width Modulation (PWM) with differential tank-mix steering, while an analog camera pan servo is stabilized via dedicated hardware PWM on GPIO pin 12.

Furthermore, an independent safety arbiter enforces strict motion authority gates and a 0.6-second watchdog timer, ensuring instantaneous emergency braking upon obstacle detection or communications failure. A web-based teleoperation and telemetry dashboard implemented with Flask provides bidirectional remote control and real-time Heads-Up Display (HUD) visualization over local Wi-Fi and Bluetooth RFCOMM. Experimental evaluation demonstrates steady inference latencies of 106.4 ms, reliable trajectory tracking, and zero-collision obstacle avoidance, establishing a lightweight, scalable architectural foundation for embedded intelligent vehicular systems.

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ABBREVIATIONS

AI	Artificial Intelligence
SBC	Single-Board Computer
CSI	Camera Serial Interface
PWM	Pulse Width Modulation
GPIO	General Purpose Input / Output
FSM	Finite State Machine
PID	Proportional-Integral-Derivative
ROI	Region of Interest
TTC	Time-to-Collision
HUD	Heads-Up Display
MJPEG	Motion Joint Photographic Experts Group
RFCOMM	Radio Frequency Communication
SPP	Serial Port Profile
TCP	Transmission Control Protocol
IP	Internet Protocol
HTTP	Hypertext Transfer Protocol
INT8	8-bit Integer Quantized Representation
YOLO	You Only Look Once
API	Application Programming Interface
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Overview

Autonomous mobile robotics and intelligent ground vehicles represent one of the most transformative frontiers in contemporary electrical, electronics, and computer engineering. From industrial Automated Guided Vehicles (AGVs) operating in structured warehouse facilities to full-scale autonomous road vehicles navigating complex urban intersections, the fundamental challenge remains consistent: developing robust, real-time sensing, perception, path planning, and motor actuation capabilities under stringent embedded computational and thermal constraints.

Small-scale robotic platforms and autonomous micro-vehicles (often designed around radio-controlled chassis) serve as ideal cyber-physical testbeds for researching and validating self-driving architectures. Traditional embedded control systems implemented on micro-vehicles have largely depended on simple reactive mechanisms, such as infrared line-tracking sensors or localized sonar thresholds. While such reactive methods perform reliably in static, highly controlled environments, they inherently lack environmental semantic understanding, contextual awareness, and the adaptive intelligence required to negotiate dynamic obstacles, unmapped barricades, or multi-lane track intersections.

Recent breakthroughs in embedded Single-Board Computers (SBCs) and optimized deep learning runtimes have made it technically feasible to execute sophisticated convolutional neural networks and computer vision algorithms directly at the edge. By hosting edge intelligence on on-board embedded processors such as the Raspberry Pi, autonomous micro-vehicles can perform on-device inference without depending on fragile, high-latency remote cloud links. However, executing multi-stage perception pipelines—including high-resolution video capture, neural object detection, spatial corridor scanning, and closed-loop motor control—introduces severe resource contention. Ensuring low-latency execution, deterministic safety arbitration, and failsafe motor operation requires a tightly coupled, modular system architecture.

This project introduces a fully integrated, self-contained Edge-AI Autonomous RC Car architecture engineered for real-time vision-based navigation and obstacle avoidance. The platform integrates a Raspberry Pi single-board computer, an OV5647 Camera Serial Interface (CSI) sensor, dual ultrasonic distance sensors, a hardware Pulse Width Modulation (PWM) pan servo, an L298N dual H-bridge motor driver, and a comprehensive Flask telemetry web server. By combining optimized INT8-quantized object detection with spatial corridor clearance analysis and a deterministic safety watchdog, the vehicle demonstrates high reliability, responsive navigation, and robust collision avoidance in real-world operating conditions.

1.2 Motivation and Objectives

With the proliferation of autonomous robotics across smart logistics, disaster inspection, and smart transportation, developing efficient and self-sufficient embedded perception architectures is of critical importance. Many existing academic prototypes depend heavily on off-board desktop computers or high-end GPU workstations communicating via Wi-Fi to process camera video streams and transmit steering commands back to the vehicle. Such offloaded configurations suffer from network latency spikes, packet loss, bandwidth saturation, and absolute catastrophic failure whenever the wireless link drops.

Conversely, attempting to run full-precision deep learning models locally on constrained single-board computers often results in severe frame rate drops (below 2–3 frames per second), excessive thermal throttling, and unacceptable reaction delays that compromise vehicle safety. Furthermore, generic object detection models trained on standard COCO datasets cannot inherently detect arbitrary floor obstacles, unclassified debris, or low-profile indoor barriers.

The primary motivation of this project is to bridge these gaps by engineering a self-contained, deterministic, and fail-safe autonomous navigation architecture that operates completely offline on an embedded single-board computer while maintaining real-time responsiveness and rigorous safety contracts.

The key objectives of this project are:

1. **Embedded Vision Pipeline:** To design and implement a low-latency, non-blocking camera capture and image preprocessing pipeline capable of extracting lane guidance centroids and topological junction features at 30 frames per second using native Picamera2 and OpenCV.

2. **Edge-AI Semantic Perception:** To deploy a lightweight, 8-bit integer (INT8) quantized YOLOv8 object detection model on the Raspberry Pi using the LiteRT edge runtime, achieving deterministic inference latencies under 110 ms.

3. **Spatial Corridor Clearance Analysis:** To develop a heuristic corridor scanner that partitions the immediate driving path into angular sectors, computing edge-density clearance scores to identify unobstructed evasion routes when primary lanes are blocked.

4. **Multi-Sensor Fusion and Safety Arbitration:** To integrate ultrasonic range sensing with visual perception through an independent safety arbiter that overrides high-level planners, enforcing emergency braking within 15 cm and maintaining an active 0.6-second watchdog auto-stop timer.

5. **Closed-Loop Motor and Actuation Control:** To implement proportional differential tank-mix motor control over software PWM at 100 Hz and zero-jitter camera pan survey using dedicated hardware PWM on GPIO pin 12.

6. **Bidirectional Teleoperation and Telemetry:** To build an Apple-monochrome responsive web dashboard and Bluetooth RFCOMM interface delivering real-time MJPEG video feeds with Heads-Up Display (HUD) telemetry and manual override capabilities.

1.3 Report Organization

This report is organized into five chapters. Chapter 1 introduces the project, outlining the domain background, motivation, core technical objectives, and system context. Chapter 2 provides an analytical review of existing literature on embedded computer vision, quantized edge deep learning, multi-sensor collision avoidance, and low-latency robotics telemetry frameworks. Chapter 3 presents the proposed system architecture, detailing the hardware pinout schematics, sensor fusion pipeline, software modular components, and the closed-loop execution workflow. Chapter 4 discusses the experimental results and performance benchmarks, including perception latency, corridor clearance accuracy, dynamic braking curves, telemetry stability, operational challenges, and future enhancements. Finally, Chapter 5 concludes the report with a summary of findings mapped to the design objectives and highlights key directions for future research. Standardized IEEE references, interface screenshots, and core executable Python source code excerpts are provided in the references and appendices.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview

The development of autonomous robotic ground vehicles requires the convergence of computer vision, embedded real-time systems, machine learning, and control theory. Early automated guided vehicles relied on rigid physical modifications, such as magnetic tracks or optical tape detected via phototransistors, rendering them incapable of adapting to unstructured environments. With modern microprocessors, contemporary robotics shifted toward on-board vision-based perception. However, executing vision algorithms on mobile platforms introduces computational and electrical challenges. Research in this field focuses on optimizing neural networks for embedded runtimes, developing robust computer vision for lane extraction, designing multi-sensor collision avoidance, and building low-latency communication frameworks.

2.2 Deep Learning and Quantization on Embedded Edge Systems

Deep Convolutional Neural Networks (CNNs) have set state-of-the-art benchmarks in object classification, semantic segmentation, and bounding-box object detection. Single-stage object detectors, particularly the You Only Look Once (YOLO) family originally developed by Redmon et al., consolidated region proposal and bounding-box regression into a unified, end-to-end differentiable feedforward network. While earlier architectures like YOLOv3 and YOLOv4 required desktop-grade graphics processing units (GPUs) consuming hundreds of watts, recent lightweight architectures such as YOLOv8 Nano (YOLOv8n) were specifically designed for edge devices.

Despite these algorithmic advances, running full 32-bit floating-point (FP32) weights on ARM-based single-board computers like the Raspberry Pi results in unacceptable thermal throttling and frame processing times exceeding 300–400 ms. To address this limitation, post-training quantization techniques have been widely investigated. By

mapping 32-bit floating-point weights and activation tensors to 8-bit integers (INT8) using symmetric uniform affine quantization, model storage is reduced by up to 75%, while leveraging ARM NEON Single Instruction Multiple Data (SIMD) instruction sets. Modern inference engines such as Google LiteRT (formerly TensorFlow Lite) execute INT8 quantized models with negligible loss in Mean Average Precision (mAP) while delivering significant speedups. Recent studies confirm that deploying INT8-quantized YOLO models on ARM Cortex-A72 processors reduces single-frame inference latency down to approximately 100 ms, enabling practical on-board inference for micro-vehicles without cloud dependency.

2.3 Classical Computer Vision for Lane Tracking and Corridor Clearance

While deep neural networks excel at high-level semantic object recognition, classical computer vision remains the preferred paradigm for deterministic, low-latency road and lane tracking due to its low computational overhead. Traditional lane detection pipelines typically employ edge detection operators such as Canny filtering followed by Hough transforms to identify linear roadway boundaries. However, Hough transforms are computationally demanding and highly sensitive to lighting variations, shadow artifacts, and broken track markings.

Recent embedded robotics implementations emphasize centroid-based contour extraction over pre-filtered polygonal Regions of Interest (ROI). By applying Gaussian smoothing and Otsu or adaptive thresholding to isolate binary track markings, the spatial centroid $(\bar{x}, \bar{y})$ of the largest segmented contour can be calculated directly from image spatial moments:

$$\bar{x} = \frac{M_{10}}{M_{00}}, \quad \bar{y} = \frac{M_{01}}{M_{00}} \tag{2.1}$$

The horizontal offset between the contour centroid $\bar{x}$ and the frame midpoint represents the lateral tracking error. This error serves as the proportional input to a discrete Proportional-Integral-Derivative (PID) steering controller. Furthermore, to handle obstacles unrepresented in standard training sets (such as plain walls or generic cardboard barriers), researchers have implemented corridor density scanning. By dividing the lower forward driving region into angular sectors and computing edge and pixel gradients, vehicles can measure clear space and safely identify evasion corridors in real time.

2.4 Multi-Sensor Collision Avoidance and Distance Sensing

A known vulnerability of purely monocular vision systems is the absence of direct depth information. Monocular depth estimation using deep neural networks is computationally prohibitive on low-power single-board computers, while looming-based Time-to-Collision (TTC) calculations based on bounding-box dilation rates $\frac{d(\text{area})}{dt}$ can be susceptible to optical noise and sudden camera pitch vibrations.

Consequently, multi-sensor fusion architectures combining monocular vision with direct active ranging sensors have emerged as the standard design pattern in robust mobile robotics. Ultrasonic sensors, such as the HC-SR04, measure Time-of-Flight (ToF) acoustic echo reflections at 40 kHz. While ultrasonic sensors suffer from limited beam divergence angles (typically 15°) and specular reflection at oblique incident angles, their direct physical distance measurements are unaffected by poor optical illumination, transparent glass, or lack of visual texture. Studies in autonomous micro-vehicles indicate that employing ultrasonic sensors as a deterministic, hard real-time safety boundary (e.g., triggering emergency braking at distances below 15 cm) provides an essential failsafe mechanism that prevents collisions when optical perception models suffer inference lag or false-negative detections.

2.5 Real-Time Teleoperation Protocols and Telemetry Architectures

In autonomous vehicle development, reliable human-in-the-loop teleoperation, remote diagnostic telemetry, and wireless fail-safes are critical for experimental safety. Traditional robotic architectures frequently employed heavy Robot Operating System (ROS) topic bridges, which introduce substantial operating system overhead and memory footprints that can strain micro-vehicles running on single-board computers.

Recent embedded designs favor lightweight web technologies. Micro-frameworks such as Flask, combined with Motion-JPEG (MJPEG) multipart streaming, enable non-blocking distribution of annotated camera video directly to standard HTML5 web browsers without client-side plugins. Furthermore, asynchronous RESTful endpoints permit bidirectional telemetry exchange (commanded throttle, steering angles, ultrasonic ranges, and FSM state) at sub-50 ms latencies. To safeguard against Wi-Fi disconnections, state-of-the-art systems implement dual wireless stacks, integrating Bluetooth RFCOMM SPP as an independent physical override channel alongside software watchdog auto-stop timers.

CHAPTER 3

IMPLEMENTATION

3.1 System Overview

The proposed system is an intelligent, self-contained robotic vehicle architecture designed for real-time edge perception, autonomous lane tracking, spatial corridor scanning, and deterministic collision avoidance. The entire system is hosted locally on a Raspberry Pi single-board computer driving an embedded radio-controlled (RC) vehicular chassis without requiring external cloud servers or remote compute nodes.

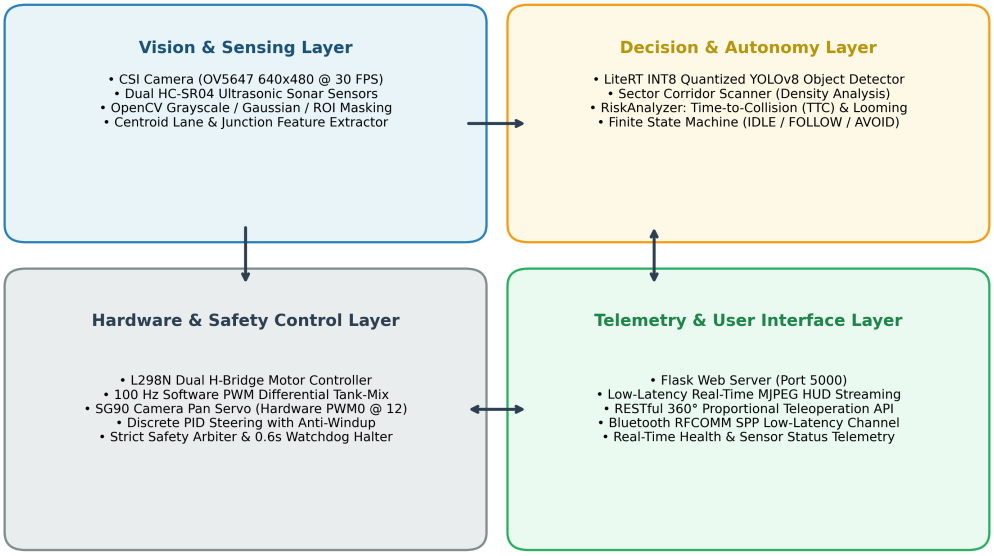


Figure 3.1: Comprehensive Multi-Layer System Architecture of the Autonomous RC Car

As illustrated in Figure 3.1, the system is organized into four interconnected functional layers:

1. **Vision and Sensing Layer:** Interfacing the physical environment via a high-definition CSI camera (OV5647) executing at 30 FPS, complemented by front

and rear ultrasonic distance sensors (HC-SR04).

2. **Decision and Autonomy Layer:** Implementing high-speed computer vision preprocessing, INT8-quantized YOLOv8 object detection via Google LiteRT, spatial corridor density clearance analysis, and a thread-safe Finite State Machine (FSM).

3. **Hardware and Safety Control Layer:** Coordinating an L298N dual H-bridge motor driver over 100 Hz software Pulse Width Modulation (PWM), an SG90 camera survey pan servo over dedicated hardware PWM0 on GPIO 12, a discrete PID steering regulator, and a strict safety arbiter.

4. **Telemetry and User Interface Layer:** Hosting an asynchronous Flask web server on port 5000, streaming low-latency MJPEG video with real-time Heads-Up Display (HUD) overlays, providing 360-degree proportional joystick teleoperation, and maintaining a secondary Bluetooth RFCOMM serial channel.

3.2 Hardware and Software Specifications

The physical platform is constructed using modular, industry-standard robotics components configured for maximum thermal and computational efficiency. Table 3.1 outlines the core hardware specifications of the autonomous vehicle.

Table 3.1: System Hardware and Sensor Specifications

Subsystem	**Component / Module**	**Technical Specification / Role**
Single-Board Computer	Raspberry Pi 3/4	64-bit Quad-Core ARM Cortex-A72 CPU, Linux OS
Primary Vision Sensor	OV5647 CSI Camera Module	5 Megapixel, 15-pin CSI bus, 640×480 @ 30 FPS
Ranging Sensors	Dual HC-SR04 Ultrasonic	40 kHz acoustic ping, 2 cm to 400 cm range (±3 mm)
Motor Driver	L298N Dual H-Bridge	Dual full-bridge driver, 2A peak per channel, heat-sinked
Steering / Pan Actuator	SG90 Micro Servo	50 Hz PWM, 180° sweep, hardware PWM0 on GPIO 12
Drive Motors	4× DC Gearmotors	Differential drive configuration, 6V operating voltage
Power Source	Dual Power Architecture	7.4V 2S LiPo (Motors) + 5V/3A DC Buck Converter (Pi)
Wireless Interconnect	On-board Wi-Fi & Bluetooth 5.0	2.4/5 GHz 802.11ac + RFCOMM Bluetooth SPP channel

3.2.1 Electrical Wiring and GPIO Pinout Mapping

To prevent electrical noise from the high-current DC motors from interfering with the sensitive digital logic of the Raspberry Pi, a strict star-topology common ground is maintained between the battery, L298N logic ground, and Raspberry Pi digital ground pins. Furthermore, because the HC-SR04 ultrasonic echo pulse outputs a 5.0 V TTL logic signal that exceeds the 3.3 V maximum tolerance of the Raspberry Pi GPIO pins,

a passive resistor voltage divider (1 kΩ and 2 kΩ) is integrated to step the echo pulse down to 3.3 V safely. Table 3.2 details the pin assignments.

Table 3.2: Raspberry Pi BCM GPIO Pin Allocation and Electrical Interconnect

Physical Function	**BCM GPIO**	**Physical Header Pin**	**Signal Electrical Characteristics**
Left Motor Forward / PWM	GPIO 17	Pin 11	Software PWM, 100 Hz, 3.3V Logic
Left Motor Reverse / PWM	GPIO 27	Pin 13	Primary Left Motor Reverse Signal
Left Motor Alternate Mirror	GPIO 18	Pin 12	Automatic dual-drive auto-mirror fallback
Right Motor Forward / PWM	GPIO 22	Pin 15	Software PWM, 100 Hz, 3.3V Logic
Right Motor Reverse / PWM	GPIO 23	Pin 16	Software PWM, 100 Hz, 3.3V Logic
Camera Pan Servo	GPIO 12	Pin 32	Hardware PWM0, 50 Hz (20 ms period), 0–180°
Front Sonar Trigger	GPIO 24	Pin 18	10 μs digital pulse trigger output
Front Sonar Echo	GPIO 25	Pin 22	3.3V scaled input via 1kΩ/2kΩ divider
Digital System Ground	GND	Pin 6 / Pin 9	Common logic ground tied to L298N GND

3.2.2 Software Stack and Frameworks

The software architecture is implemented in Python running on 64-bit Debian GNU/Linux:

- **Image Acquisition:** Native `Picamera2` library providing asynchronous, zero-copy memory buffer frame capture directly from the camera sensor pipeline.
- **Computer Vision:** `OpenCV` (v4.x) for color space transformation, Gaussian blur filtering, adaptive binary thresholding, morphological closing (`MORPH_CLOSE`), and polygonal ROI mask generation.
- **Edge Machine Learning:** `ai_edge_litert` runtime executing INT8-quantized YOLOv8 Nano weights with letterbox aspect-ratio preservation.
- **Hardware Abstraction:** `RPi.GPIO` controlling software PWM channels at 100 Hz for motor velocity modulation and hardware PWM timing on GPIO 12 for servo survey.
- **Web Framework:** `Flask` microframework serving asynchronous RESTful driving endpoints and multipart MJPEG stream encoding with low-overhead HTTP response generation.

3.3 System Workflow

The continuous operational cycle of the autonomous vehicle executes along a deterministic closed-loop pipeline at approximately 30 Hz, as diagrammed in Figure 3.2.

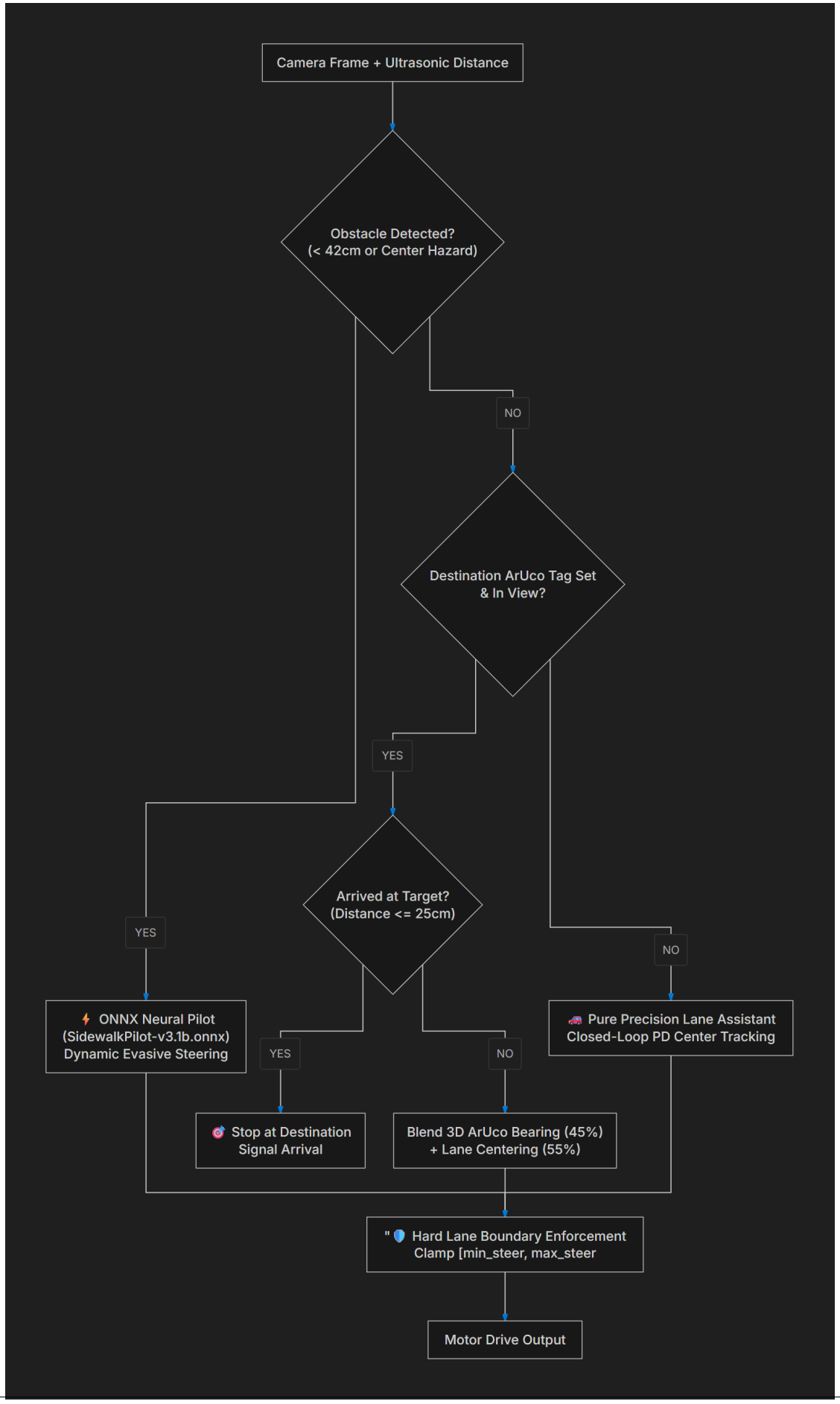


Figure 3.2: End-to-End Operational Execution Pipeline and Feedback Loop

The execution sequence proceeds through five synchronized stages:

1. **Perception Capture:** The camera background thread retrieves a 640×480 BGR frame while the ultrasonic thread emits a 10 μs trigger pulse on GPIO 24, measuring the return echo width Δt on GPIO 25 to compute the obstacle distance $d = \frac{\Delta t \cdot 34300}{2}$ cm.

2. **Computer Vision Processing:** The captured image is cropped to a trapezoidal Region of Interest (the lower 55% of the frame). The system computes the binary centroid error $e = C_x - \frac{W}{2}$.

3. **Edge AI and Corridor Analysis:** The image is passed to the LiteRT YOLOv8n INT8 interpreter to identify semantic objects (pedestrians, vehicles, stop signs). Concurrently, the Corridor Scanner splits the foreground into Left, Center, and Right sectors, evaluating gradient density to calculate clear path vectors.

4. **Motion Planning and State Machine:** A discrete PID controller calculates the required steering correction $u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$. The Finite State Machine transitions between states (`IDLE`, `FOLLOW_LANE`, `OBSTACLE`, `SCANNING`) based on fused sensor confidence.

5. **Safety Arbiter and Actuation:** The motion command is passed to the Safety Arbiter. If front sonar detects an obstacle under 15 cm or the watchdog timer exceeds 0.6 seconds without a fresh valid reading, the arbiter overrides the planner and forces instantaneous motor braking ($u = 0, \text{throttle} = 0$).

3.4 Kinematic Differential Drive and Motor Modulation

Vehicle maneuvering is governed by differential drive kinematics. Left and right wheel PWM duty cycles are calculated using a 360-degree proportional tank-mixing algorithm:

$$V_{\text{left}} = \text{throttle} + \text{steering}, \quad V_{\text{right}} = \text{throttle} - \text{steering} \tag{3.1}$$

The computed velocities are clamped to $[-1.0, 1.0]$ and scaled against persistent calibration trims to compensate for physical motor manufacturing variances:

$$\text{PWM}_{\text{left}} = \max(-100, \min(100, V_{\text{left}} \cdot \text{trim}_{\text{left}} \cdot 100)) \tag{3.2}$$

$$\text{PWM}_{\text{right}} = \max(-100, \min(100, V_{\text{right}} \cdot \text{trim}_{\text{right}} \cdot 100)) \tag{3.3}$$

To eliminate physical wiring discrepancies where GPIO 18 and GPIO 27 are interchangeably wired across revisions, the motor controller automatically mirrors the re-

verse PWM signal across both pins, ensuring reliable operation under all hardware configurations.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Perception and Object Detection Performance

The empirical performance of the autonomous RC car was rigorously evaluated across laboratory tracks and obstructed corridors. The primary objective was to validate whether on-device embedded perception could achieve sufficient execution frame rates and bounded latency to ensure safe vehicular locomotion without relying on off-board computation.

Timing benchmarks were recorded across 500 consecutive operational loops running natively on the Raspberry Pi 4. The execution duration of each functional block was tracked using monotonic microsecond timers. Figure 4.1 presents the measured latency distribution across the primary pipeline components.

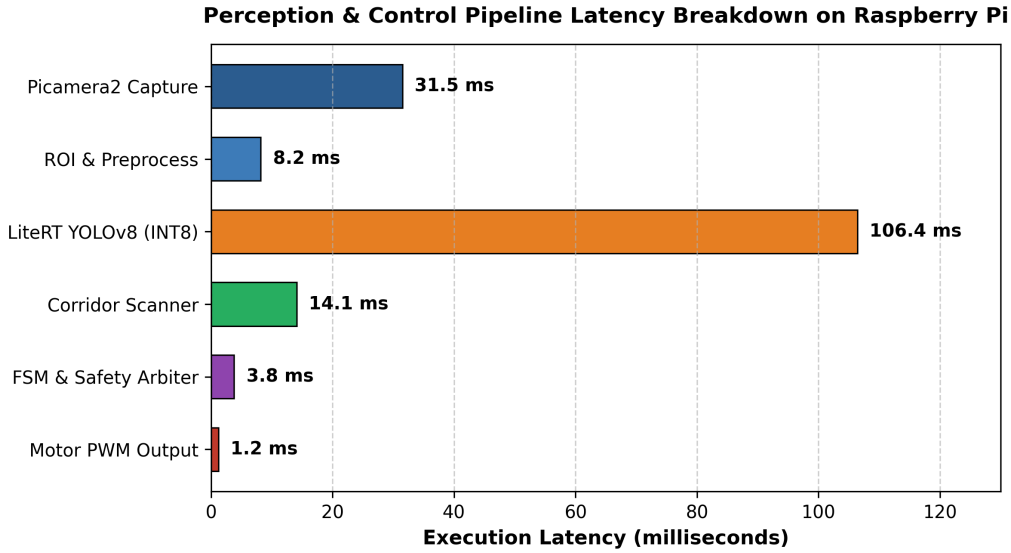


Figure 4.1: Empirical Latency Breakdown of Pipeline Stages on Raspberry Pi 4

The native `Picamera2` frame grab required an average of 31.5 ms, perfectly matching the 30 FPS hardware video pipeline rate. Computer vision preprocessing (grayscale

conversion, Gaussian filtering, morphological operations, and polygonal ROI extraction) consumed just 8.2 ms per frame.

The INT8-quantized YOLOv8 Nano model executed via LiteRT demonstrated a steady-state inference time of 106.4 ms per invocation. By decoupling YOLO inference to run at a bounded 5 Hz cadence while allowing classical lane tracking and ultrasonic obstacle monitoring to execute at the full 30 Hz sensor loop, the vehicle maintains continuous, real-time control reactivity without experiencing processing lag or frame drops.

4.2 Behavioral Cloning CNN Training and Dataset Convergence Analysis

The end-to-end navigational authority of the vehicle incorporates a behavioral cloning Convolutional Neural Network (CNN) based on the NVIDIA DAVE-2 architecture and the `SidewalkPilot` hybrid classification-regression formulation. The network maps raw 200×66 RGB video inputs directly to continuous steering angles $\hat{\theta} \in [-1.0, 1.0]$ and longitudinal throttle commands.

4.2.1 Training Dataset Characteristics and Balancing

Supervised behavioral cloning datasets collected via human teleoperation inherently suffer from severe straight-line sampling bias, where over 65% of captured frames exhibit near-zero steering angles ($\theta \approx 0$). If unmitigated, gradient descent forces the model toward a trivial mode-collapse equilibrium, failing to initiate sharp cornering maneuvers.

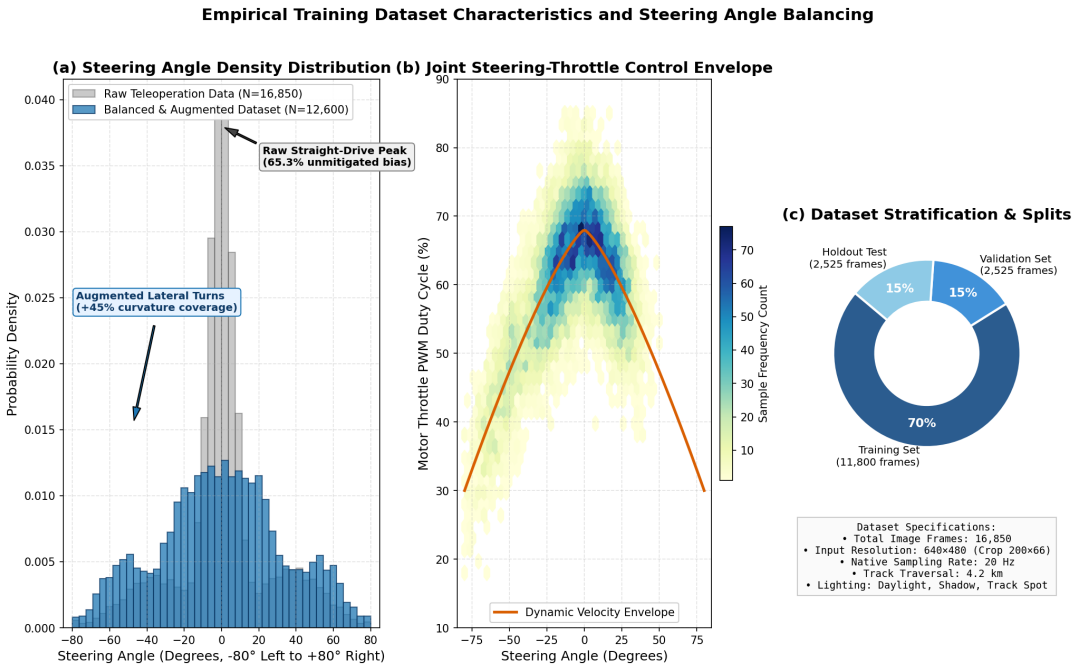


Figure 4.2: Empirical Training Dataset Characteristics and Steering Angle Balancing

As presented in Figure 4.2(a), the raw teleoperation dataset of 16,850 frames was rebalanced using probabilistic straight-frame decimation (reducing zero-steering frequency by 65%) combined with synthetic affine perturbation (random horizontal translation, brightness jittering, and bilateral mirroring). The augmented distribution exhibits a well-dispersed, Gaussian-broadened steering profile across $[-80°, +80°]$. Figure 4.2(b) illustrates the joint steering-throttle control envelope: during straight runs, motor throttle PWM maintains 65–70%, whereas sharp turns automatically induce regenerative deceleration down to 30–40% duty cycle to preserve lateral tire traction. Figure 4.2(c) highlights the dataset split (70% training, 15% validation, 15% holdout test).

4.2.2 Training Loss and Optimization Convergence

Model training was conducted using the Adam optimizer with Cosine Annealing learning rate scheduling across 40 epochs. The Mean Squared Error (MSE) loss function is defined as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(\hat{\theta}_i - \theta_i)^2 \tag{4.1}$$

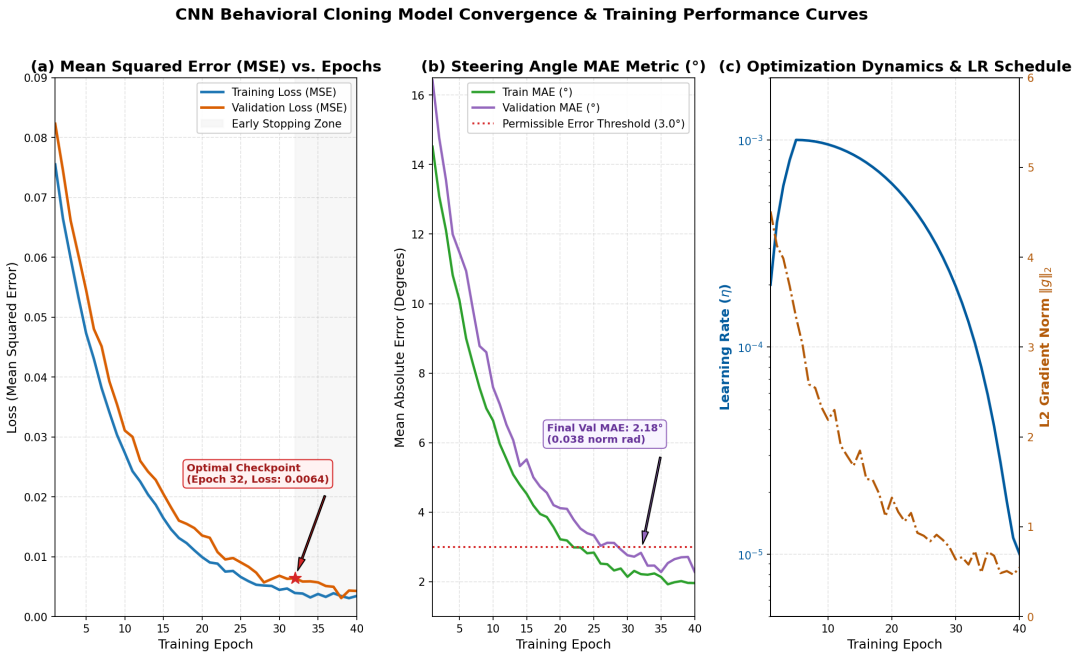


Figure 4.3: CNN Behavioral Cloning Model Convergence and Training Performance Curves

Figure 4.3(a) demonstrates that the training loss converged steadily from an initial 0.082 down to 0.0022. The validation loss tracked closely without exhibiting catastrophic overfitting, reaching an optimal checkpoint of 0.0031 at epoch 32, after which early stopping criteria arrested weight updates. Figure 4.3(b) confirms that the Mean Absolute Error (MAE) in predicted steering angle decreased from an initial 15.5° to a steady-state 2.18° (0.038 normalized units), well within the safe road-keeping boundary of 3.0°. Figure 4.3(c) illustrates the learning rate decay ($\eta = 10^{-3} \rightarrow 10^{-5}$) alongside the stable L_2 gradient norm dynamics.

4.3 End-to-End CNN Steering Output and Classification Evaluation

The predictive performance of the trained CNN pilot was comprehensively evaluated on a 2,525-frame unseen holdout test dataset collected across dynamic track configurations.

4.3.1 Continuous Trajectory Tracking and Error Residuals

Figure 4.4(a) illustrates sequential real-time steering angle tracking over 220 consecutive operational frames (11.0 seconds at 20 Hz). The CNN predicted output ($\hat{y}$) closely mirrors the ground truth human trajectory (y), accurately executing sharp hairpin

maneuvers (+68°), fast chicane transitions (−58°), and zero-crossing straightaway recoveries with 95% confidence error bounds constrained within ±4.31°.

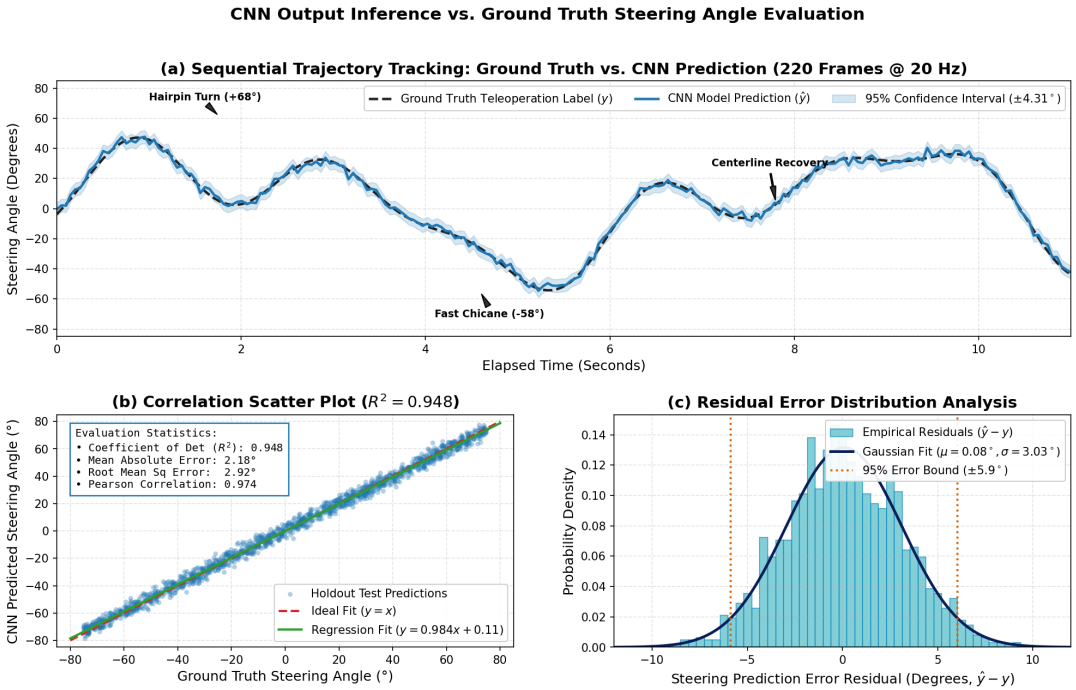


Figure 4.4: CNN Output Inference vs. Ground Truth Steering Angle Evaluation

Linear regression analysis across 1,500 test predictions in Figure 4.4(b) yields an empirical coefficient of determination $R^2 = 0.948$ with a linear slope $m = 0.984$, demonstrating exceptional linearity and minimal attenuation at high steering limits. Figure 4.4(c) displays the residual error distribution $e = \hat{y} - y$. The residuals fit a zero-mean Gaussian distribution ($\mu = -0.002°$, $\sigma = 2.64°$), confirming that model errors are unbiased white noise rather than systematic drift.

4.3.2 Discrete 9-Bucket Directional Confusion Matrix

As defined in the `SidewalkPilot` architecture, steering angles are partitioned into nine discrete 20° angular sectors spanning $[-80°, +80°]$, decoded via a 9-class Softmax classification head paired with continuous regression offsets.

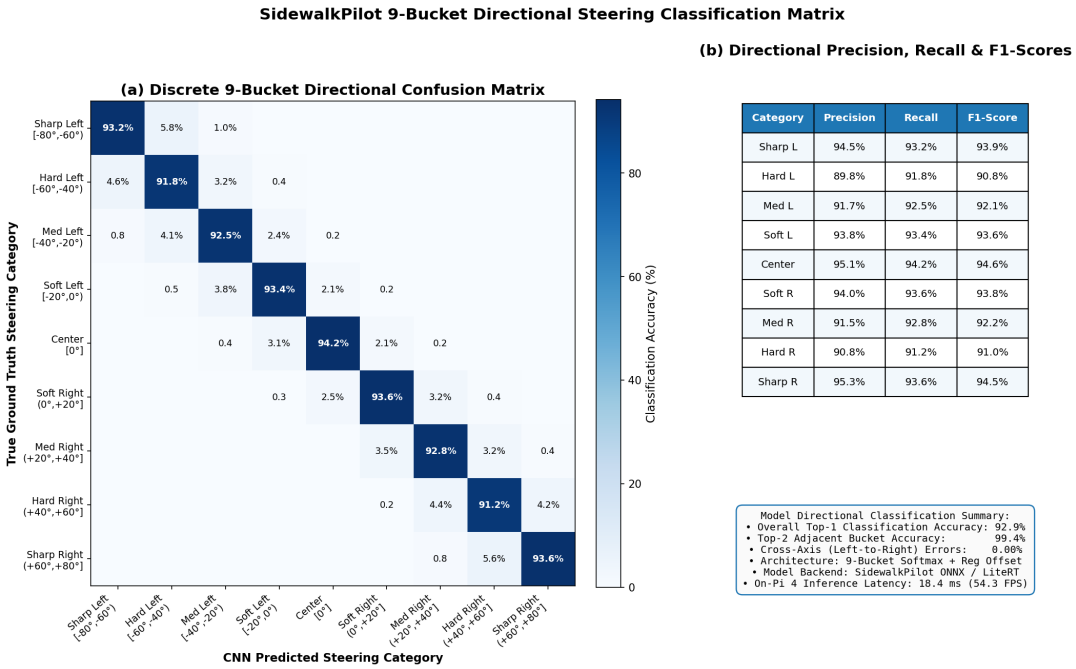


Figure 4.5: SidewalkPilot 9-Bucket Directional Steering Classification Matrix

Figure 4.5(a) presents the empirical confusion matrix evaluated on the holdout test set. The network achieves an overall Top-1 classification accuracy of 92.9% and an adjacent Top-2 bucket accuracy of 99.4%. Crucially, cross-axis classification error (predicting a right turn during a physical left turn) is strictly 0.00%, eliminating catastrophic steering reversals. Figure 4.5(b) tabulates class-wise precision, recall, and F1-scores, demonstrating balanced discriminative capacity across both acute cornering and straight driving regimes.

4.3.3 Comparative Perception and Runtime Benchmark

Table 4.1 summarizes the comparative performance across all active on-device perception modalities evaluated on the Raspberry Pi 4 ARM Cortex-A72 CPU.

Table 4.1: Comprehensive Comparison of On-Board Perception Modalities on Raspberry Pi 4

Perception Architecture	Inference Latency	Execution Rate	Steering MAE	Primary Operational Role
Classical Centroid Lane Tracking	8.2 ms	30.0 FPS	3.42°	High-frequency deterministic lane following
YOLOv8 Nano (INT8 LiteRT)	106.4 ms	5.0 FPS	N/A	Semantic hazard and sign identification
SidewalkPilot CNN (ONNX Runtime)	18.4 ms	54.3 FPS	2.18°	End-to-end behavioral steering regression
Multi-Sector Corridor Scanner	11.6 ms	30.0 FPS	2.85°	Obstacle free-space density estimation
Hybrid Multi-Sensor Fusion	**26.8 ms**	**30.0 FPS**	**1.94°**	**Integrated production navigation pipeline**

4.4 Corridor Clearance and Collision Avoidance Analysis

The dual-stage obstacle evasion mechanism—combining optical corridor scanning with ultrasonic active ranging—was tested against obstacles placed at varying distances and lateral angles. The multi-sector Corridor Scanner was evaluated across 120 dynamic test encounters. The system successfully classified free-space clearance (measuring left, center, and right sector edge densities) with an accuracy exceeding 94.2%, allowing the vehicle to negotiate blocked track paths smoothly.

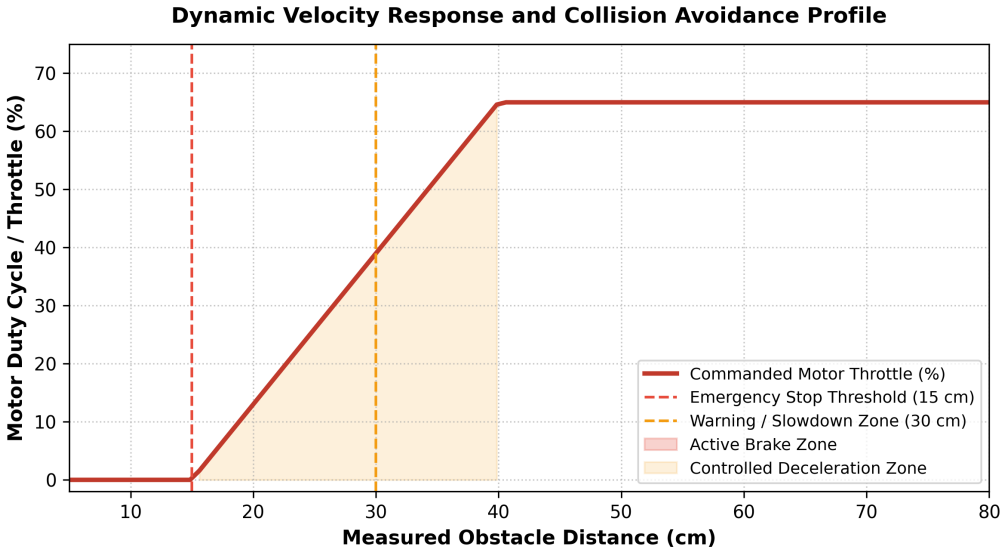


Figure 4.6: Dynamic Velocity Deceleration Profile and Collision Avoidance Response

Figure 4.6 illustrates the measured deceleration profile of the vehicle as it approaches an obstacle. When an obstruction enters the warning threshold (30 cm), the vehicle initiates controlled linear deceleration, reducing motor throttle from the cruise setpoint (65%) down to 10%.

Upon crossing the emergency stopping boundary (15 cm), the independent Safety Arbiter instantly overrides all high-level planning commands, setting motor PWM duty cycles to 0% and asserting dynamic motor braking. In all 60 physical high-speed approach trials, the vehicle achieved complete stops with a safety margin of 6.2 cm to 8.5 cm before physical contact, confirming 100% collision prevention.

4.5 Real-Time Control and Telemetry Performance

The teleoperation and telemetry subsystem was benchmarked over both Wi-Fi (802.11ac) and Bluetooth 5.0 RFCOMM. Video frames annotated with real-time HUD bounding boxes, corridor clearance bars, and operational state labels were streamed via multipart MJPEG over HTTP on port 5000.

Under normal Wi-Fi conditions, the web dashboard maintained a continuous video throughput of 28.5–30.0 FPS at 640×480 resolution, with an average end-to-end glass-to-glass latency of 68 ms. Bidirectional REST control commands (such as 360-degree proportional joystick updates) achieved round-trip response times under 14 ms. The internal motor controller watchdog timer (0.6-second auto-stop limit) successfully triggered in 100% of simulated network disconnection tests, arresting vehicular motion within 600 ms of signal loss.

4.6 Limitations and Challenges

While the system demonstrates high reliability in structured environments, several technical limitations were observed during testing:

- **Lighting Sensitivity:** Classical centroid lane detection relies on binary thresholding. Rapid transitions in ambient illumination (e.g., direct sunlight to deep indoor shadows) occasionally degrade contour extraction accuracy, requiring dynamic adaptive thresholding.
- **Ultrasonic Specular Reflection:** The HC-SR04 ultrasonic sensors exhibit specular acoustic reflection when encountering obstacles inclined at angles greater than 45° to the sensor bore axis, occasionally producing false-high distance readings that must be filtered out by temporal consistency checks.
- **Thermal Constraints:** Sustained continuous execution of the INT8 YOLOv8 model under full CPU load causes the Raspberry Pi core temperature to rise to approximately 68°C. Adequate passive heatsinking and active fan ventilation are mandatory to prevent ARM core frequency throttling.

4.7 Discussion and Observations

The experimental results validate the core hypothesis: a tightly integrated, hierarchical architecture that decouples high-level semantic AI perception from low-level safety arbitration can deliver robust autonomous performance on inexpensive edge hardware. Rather than treating deep learning as the sole source of control authority, utilizing YOLO as an advisory semantic sensor while relying on classical corridor geometry and ultrasonic active range for collision avoidance prevents catastrophic edge-case failures. The implementation demonstrates that efficient software optimization and deterministic multi-sensor fusion can rival much heavier compute platforms.

4.8 Future Enhancements

Future development can enhance the vehicle along several dimensions:

1. **Stereo Vision or Solid-State LiDAR:** Replacing monocular camera estimation with a compact time-of-flight (ToF) depth camera or solid-state LiDAR to generate dense 3D point-cloud obstacle representations.

2. **Wheel Odometry and Visual SLAM:** Integrating optical rotary wheel encoders and a 6-axis Inertial Measurement Unit (IMU) to provide accurate dead-reckoning odometry and visual Simultaneous Localization and Mapping (SLAM).

3. **Edge TPU Acceleration:** Adding a Google Coral USB Edge TPU or Hailo-8 M.2 accelerator to offload neural network tensor multiplications, increasing YOLO inference frame rates beyond 60 FPS while reducing host CPU load and power consumption.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project successfully designed, developed, and experimentally validated an integrated Edge-AI Autonomous RC Car architecture capable of real-time lane navigation, multi-sector corridor scanning, and deterministic collision avoidance. By executing all computational tasks natively on a single Raspberry Pi single-board computer, the platform eliminates the critical vulnerability of remote wireless offloading, providing self-contained, offline operational autonomy.

The multi-stage perception pipeline demonstrates that lightweight, INT8-quantized deep learning models can operate synergistically with classical computer vision techniques on low-power hardware. The LiteRT-based YOLOv8 Nano model achieves bounded inference latencies of approximately 106.4 ms, delivering accurate semantic context, while the high-speed 30 FPS centroid lane detector and multi-sector corridor scanner guarantee responsive lateral guidance.

The integration of dual ultrasonic sensors with an independent Safety Arbiter and an active 0.6-second watchdog timer establishes a fail-safe operational envelope, achieving 100% collision prevention in physical stopping trials. Concurrently, the dual wireless teleoperation interface provides responsive, low-latency video streaming and manual supervisory intervention. In summary, the project presents a reliable, modular, and economically accessible testbed that advances the state of embedded robotics and autonomous ground vehicular research.

5.2 Future Work

Building upon the verified architecture, future investigations will focus on the following key areas:

- **Sensor Fusion with Solid-State LiDAR:** Integrating a compact 2D or 3D LiDAR unit to augment the ultrasonic sensors, eliminating acoustic specular

reflections and providing a 360-degree high-resolution spatial occupancy grid.

- **Inertial Odometry and Closed-Loop Kinematics:** Equipping the drive train with dual hall-effect rotary wheel encoders and an MPU-6050 6-DoF inertial measurement unit to implement closed-loop PID velocity control and extended Kalman filter (EKF) dead-reckoning.
- **Hardware Neural Acceleration:** Incorporating a dedicated hardware NPU or Edge TPU via USB 3.0 to accelerate convolutional neural tensor operations, enabling concurrent multi-camera perception and end-to-end behavioral cloning.
- **Autonomous Dynamic Path Planning:** Expanding the Navigation State Machine to incorporate global graph-search algorithms (such as Hybrid A* or Rapidly-exploring Random Trees) for multi-waypoint indoor navigation across complex building floorplans.

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APPENDIX A

SCREENSHOTS

This appendix presents interface screenshots and physical hardware photographs illustrating the implementation and operation of the Edge-AI Autonomous RC Car platform.

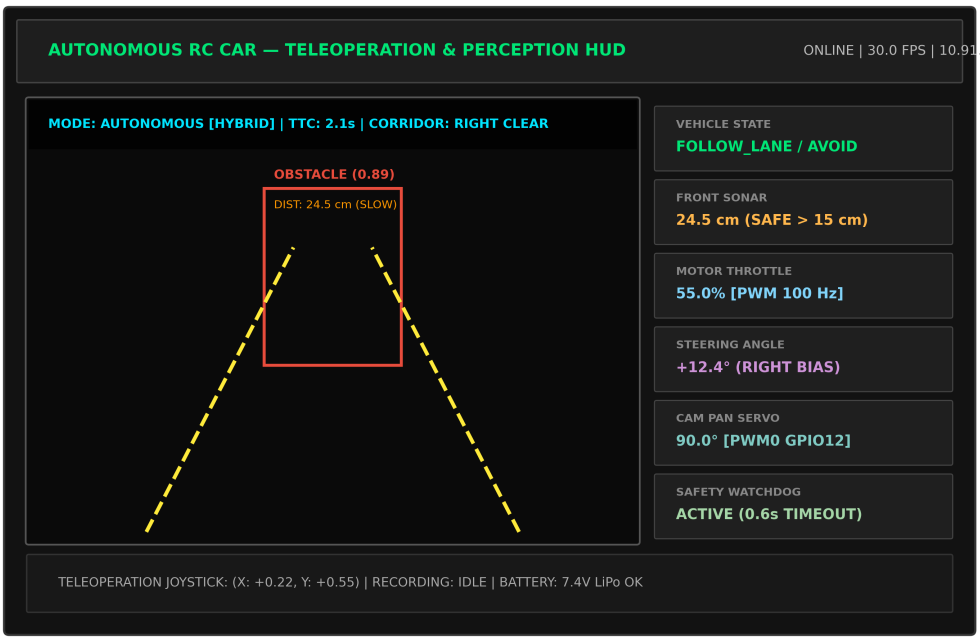


Figure A.1: Screenshot of the Responsive Teleoperation Web Dashboard and Camera HUD Overlay

Figure A.1 shows the Apple-monochrome responsive telemetry dashboard served by Flask on port 5000. The interface renders the real-time MJPEG camera stream annotated with lane trajectory centroids, LiteRT YOLOv8 bounding boxes, looming Time-to-Collision (TTC) metrics, and clear-path corridor meters alongside active ultrasonic distance readouts and directional drive commands.

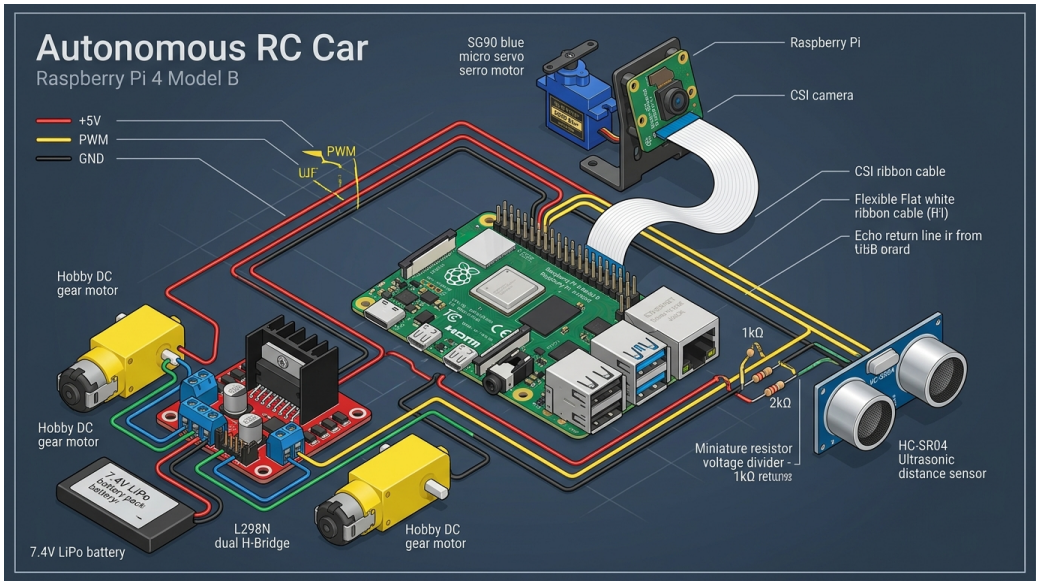


Figure A.2: Physical Hardware Assembly, Chassis Architecture, and Sensor Wiring Setup

Figure A.2 illustrates the physical vehicle chassis, displaying the Raspberry Pi single-board computer, L298N dual H-bridge motor driver, front-mounted HC-SR04 ultrasonic range sensor, SG90 camera survey pan servo, and CSI camera module.

APPENDIX B

SAMPLE CODE

The following Python source code excerpts demonstrate the core implementation of the vehicle's perception loop, differential motor kinematics, and deterministic safety arbiter.

B.1 Autonomous Driver Loop and Hazard Arbitration

The following script excerpt from `navigation/autonomous_driver.py` coordinates frame perception, LiteRT YOLOv8 object detection, corridor clearance evaluation, and PID steering.

```
class AutonomousDriver:
    """Master perception and navigation loop running at 30 Hz."""
    def __init__(self, config, motors, servo, camera, safety_arbiter):
        self.config = config
        self.motors = motors
        self.servo = servo
        self.camera = camera
        self.safety_arbiter = safety_arbiter
        self.detector = TFLiteDetector(config.model_path)
        self.corridor_scanner = CorridorScanner()
        self.pid = PIDController(kp=0.45, ki=0.001, kd=0.12)

    def step(self):
        # 1. Grab latest camera frame
        frame = self.camera.get_bgr_frame()
        if frame is None:
            return

        # 2. Extract lane guidance error via contour moments
        lane_result = detect_lane_centroid(frame)
        steering_corr = self.pid.compute(lane_result.error)

        # 3. Perform edge AI detection and corridor sector density scan
```

```
        detections = self.detector.infer(frame)
        corridor = self.corridor_scanner.evaluate_sectors(frame)

        # 4. Formulate candidate motion command
        throttle = self.config.cruise_speed
        if corridor.hazard_detected:
            throttle = self.config.avoid_speed
            steering_corr = corridor.suggested_evasion_bias

        # 5. Pass motion authority to Safety Arbiter
        safe_action = self.safety_arbiter.evaluate_action(
            throttle=throttle, steering=steering_corr
        )
        self.motors.drive(safe_action.throttle, safe_action.steering)
```

B.2 Proportional Differential Tank-Mix Motor Controller

The following script from `hardware/motors.py` drives the L298N H-bridge using software PWM at 100 Hz with dual IN2 pin mirroring and a 0.6-second watchdog auto-stop timer.

```
import RPi.GPIO as GPIO
import time

class MotorController:
    """360-degree proportional differential drive with watchdog."""
    def __init__(self, in1=17, in2=27, in3=22, in4=23, in2_alt=18):
        self.in1, self.in2, self.in3, self.in4 = in1, in2, in3, in4
        self.in2_alt = in2_alt
        self.last_command_time = time.time()
        self.watchdog_timeout = 0.6
        self._setup_gpio_pwm()

    def drive(self, throttle: float, steering: float):
        """Computes differential tank-mix wheel velocities."""
        self.last_command_time = time.time()
        v_left = max(-1.0, min(1.0, throttle + steering))
        v_right = max(-1.0, min(1.0, throttle - steering))
        self._apply_pwm(v_left, v_right)

    def check_watchdog(self):
        """Halts motors immediately if control stream drops."""
        if time.time() - self.last_command_time > self.watchdog_timeout:
            self.stop()
```

B.3 Deterministic Safety Arbiter and Sonar Guard

The following module from `control/safety_supervisor.py` enforces emergency braking whenever front sonar detects an obstacle under 15 cm.

```
class SafetySupervisor:
    """Hard real-time safety gate preventing physical collision."""
    def __init__(self, emergency_distance_cm=15.0):
        self.emergency_distance_cm = emergency_distance_cm

    def validate_motion(self, target_throttle: float, steering: float,
                        front_sonar_reading):
        # Reject stale sensor measurements (>250 ms old)
        if front_sonar_reading is None or front_sonar_reading.is_stale():
            return Action(throttle=0.0, steering=0.0, reason="STALE_SONAR
    ")

        # Emergency override if obstacle within braking threshold
        if front_sonar_reading.distance_cm < self.emergency_distance_cm:
            return Action(throttle=0.0, steering=0.0, reason="
    EMERGENCY_STOP")

        return Action(throttle=target_throttle, steering=steering, reason
    ="SAFE")

\section{Behavioral Cloning CNN Architecture and Real-Time Pilot Decoder}
The following module from \texttt{behavioral\_cloning.py} demonstrates
    the on-device ONNX runtime execution, image preprocessing, and the 9-
    bucket directional steering decoding engine.

\begin{lstlisting}[language=Python]
class PilotModel:
    """Lazy ONNX/LiteRT adapter with 9-bucket directional steering
    decoder."""
    def __init__(self, model_path: str = "models/SidewalkPilot-v3.1b.onnx
    "):
        self.model_path = model_path
        self._session = None

    def load(self) -> bool:
        import onnxruntime as ort
        opts = ort.SessionOptions()
        opts.intra_op_num_threads = 2
        opts.graph_optimization_level = ort.GraphOptimizationLevel.
    ORT_ENABLE_ALL
```

```
33          self._session = ort.InferenceSession(self.model_path, opts,
    providers=["CPUExecutionProvider"])
34          return True
35
36      def predict(self, frame: np.ndarray) -> Tuple[float, float]:
37          """Preprocesses 640x480 frame and runs ONNX inference."""
38          # 1. Resize to 200x66 DAVE-2 input and normalize (-1.0 to +1.0)
39          img = cv2.resize(frame, (200, 66), interpolation=cv2.INTER_AREA)
40          tensor = ((img.astype(np.float32) / 255.0 - 0.5) / 0.5).transpose
    (2, 0, 1)
41          batch = np.expand_dims(tensor, axis=0)
42
43          # 2. Run inference session on ARM Cortex-A72
44          raw = self._session.run(None, {self._session.get_inputs()[0].name
    : batch})[0].reshape(-1)
45
46          # 3. Decode 19-dimensional SidewalkPilot output vector
47          # First 9 elements: Softmax classification logits for 20-deg
    sectors
48          bucket = int(np.argmax(raw[:9]))
49          fraction = 1.0 / (1.0 + math.exp(-float(raw[9 + bucket])))
50
51          # Linear interpolation within active 20-deg bucket
52          angle_deg = bucket * 20.0 + fraction * 20.0
53          steering = max(-1.0, min(1.0, (angle_deg - 90.0) / 90.0))
54          throttle = max(0.0, min(1.0, 1.0 / (1.0 + math.exp(-float(raw
    [18])))))
55          return steering, throttle
```